

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – [Fitness Buddy – AI-Powered Personal Health & Fitness Coach]

Domain of Project – [Artificial Intelligence (AI) / Machine Learning (ML)]

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Recommended font style and size:

Font Style: Arial

Font Size (Heading): 28

Font Size (Content): 20

Problem Statement -

Problem Statement : Fitness Buddy

The Challenge

In today's fast-paced world, many individuals struggle to maintain a healthy lifestyle due to a lack of personalized guidance, time constraints, and inconsistent motivation. Traditional fitness solutions often require expensive subscriptions, in-person consultations, or rigid schedules that do not adapt to personal preferences or daily routines.

The Objective

Build an **AI-powered Fitness Buddy** that provides personalized fitness and healthy-lifestyle guidance based on individual needs and preferences. The solution should:

- Personalized Workout Plans** – Recommend home workouts and exercise routines based on fitness goals, fitness level, available time, and equipment.
- AI Fitness Guidance** – Provide conversational, personalized fitness advice and answer users' health and workout-related questions.
- Healthy Meal Suggestions** – Suggest simple and nutritious meal ideas based on user preferences and fitness goals.
- Daily Fitness Motivation** – Provide motivational tips and daily fitness inspiration to encourage users to stay active.
- Habit Building & Consistency** – Help users build healthy habits and maintain consistency through daily goals, reminders, and progress tracking.
- Personalized Recommendations** – Adapt workout and lifestyle suggestions according to user input and preferences.
- Progress Dashboard** – Show workout completion, habit progress, consistency, and fitness activity in an easy-to-understand dashboard.
- Workout Tracking** – Allow users to track completed workouts and view their workout history.
- Fitness Safety Guidance** – Provide basic safety recommendations and encourage users to consult qualified professionals for medical concerns.

Proposed Solution -

Proposed Solution: AI-Powered Fitness Buddy

The proposed solution is a **multi-feature AI-powered Fitness Buddy** built using **IBM Bob, IBM watsonx.ai, and IBM Granite Models** to provide personalized fitness guidance, workout recommendations, nutrition suggestions, and daily motivation.

The system uses an intelligent conversational AI assistant that understands user inputs and provides recommendations based on factors such as **fitness goal, fitness level, available time, workout location, equipment, and dietary preferences**.

The system consists of four key components:

AI Fitness Coach recommends personalized home workouts and fitness routines based on user requirements.

Workout Recommendation Module generates suitable exercises, sets, repetitions, duration, rest time, and difficulty level according to the user's fitness level and available equipment.

Nutrition & Lifestyle Module suggests simple and nutritious meals, hydration tips, healthy habits, and lifestyle recommendations based on the user's goals and preferences.

Motivation & Habit Tracking Module provides daily motivational messages, fitness tips, habit-building guidance, and tracks workout consistency and completed habits.

The solution includes a **user-friendly dashboard** that displays today's workout, daily goals, habit streaks, workout history, and overall fitness progress.

IBM Bob is used to develop and orchestrate the application workflow, while **IBM watsonx.ai** provides the AI development and deployment environment. **IBM Granite models** power the conversational assistant, reasoning, personalization, and generation of fitness and nutrition recommendations.

Overall, this solution delivers an **end-to-end intelligent fitness assistant** that helps users maintain a healthier lifestyle through personalized workouts, basic nutrition guidance, motivation, habit building, and progress tracking.

Technology Used

IBM Bob platform – For designing and orchestrating the multi-agent workflows visually.

IBM Granite model – For natural language understanding, reasoning, and personalization.

Example : ibm-granite-3-2-8b

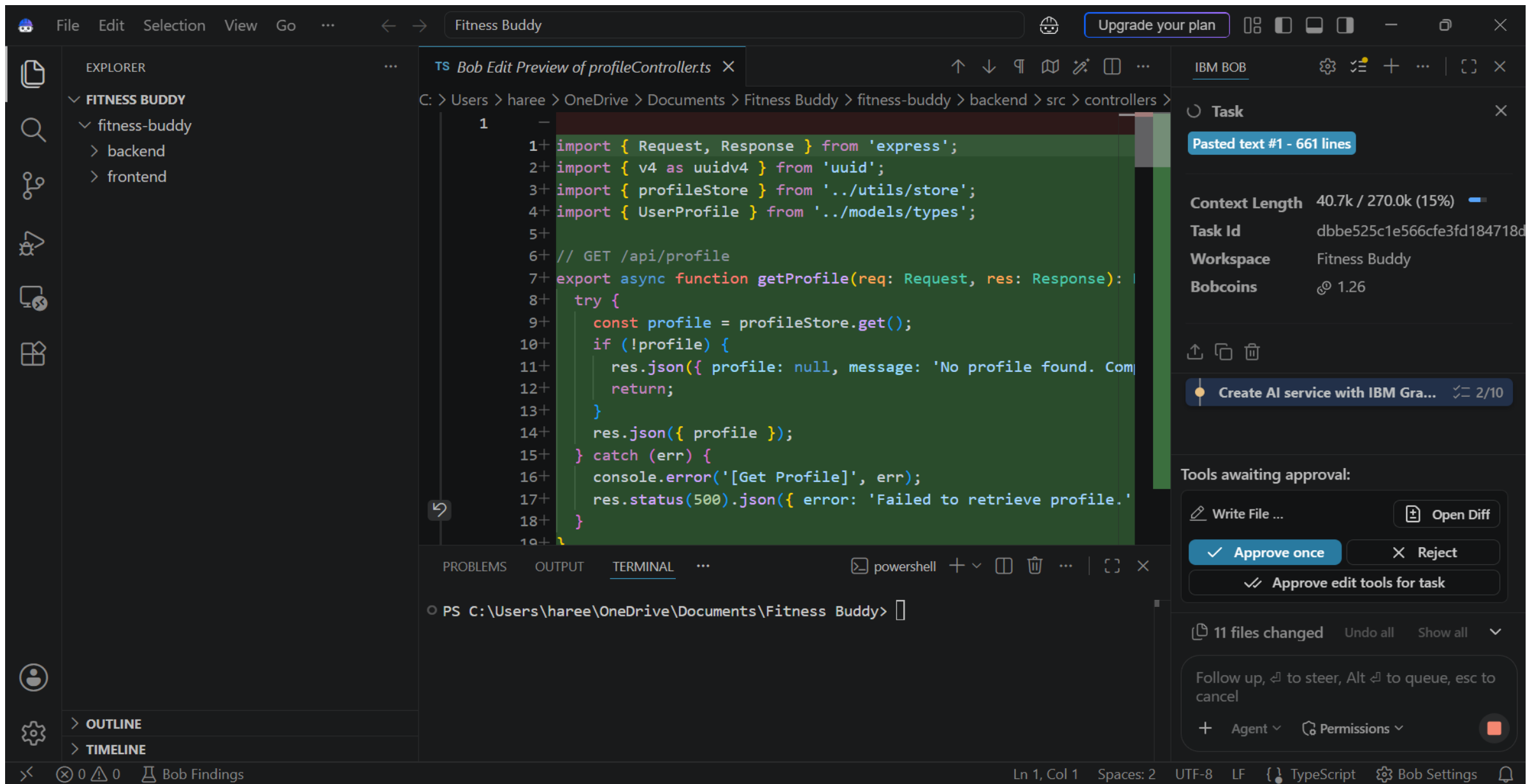
IBM Cloud – Provides the infrastructure to build, deploy, and scale the Fitness Buddy application efficiently, with a reliable and secure environment.

RAG (Retrieval-Augmented Generation) – Used to retrieve relevant and reliable fitness and nutrition information first, then generate intelligent and personalized responses.

IBM watsonx.ai – For IBM Granite model deployment, embeddings, and AI governance.

Vector Database (FAISS/Chroma) – For storing embeddings and retrieving relevant fitness and nutrition information for the RAG-based AI system.

Required Files created using IBM bob -



IBM Bob Screenshots:

File

Edit

Selection

View

Go

...

←

→

Fitness Buddy

Upgrade your plan

EXPLORER

FITNESS BUDDY

fitness-buddy

backend

frontend

node_modules

public

favicon.svg

icons.svg

src

.env.example

.gitignore

.oxlintrc.json

index.html

package-lock.json

package.json

README.md

tsconfig.app.json

tsconfig.json

tsconfig.node.json

vite.config.ts

README.md

PROBLEMS

OUTPUT

TERMINAL

...

powershell

+

✓

🗑

...

|

🔍

✕

IBM BOB

Task

Pasted text #1 - 661 lines

Context Length 104.9k / 270.0k (39%)

Task Id dbbe525c1e566cfe3fd184718d

Workspace Fitness Buddy

Bobcoins 7.01

Final validation and quality check 6/7

Both compile cleanly. Now let's do a quick build check on the frontend:

Executing cd fitness-buddy/fron...

Executing

38 files changed Undo all Show all

Follow up, ⌘ to steer, Alt ⌘ to queue, esc to cancel

Agent Permissions

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Spaces: 4

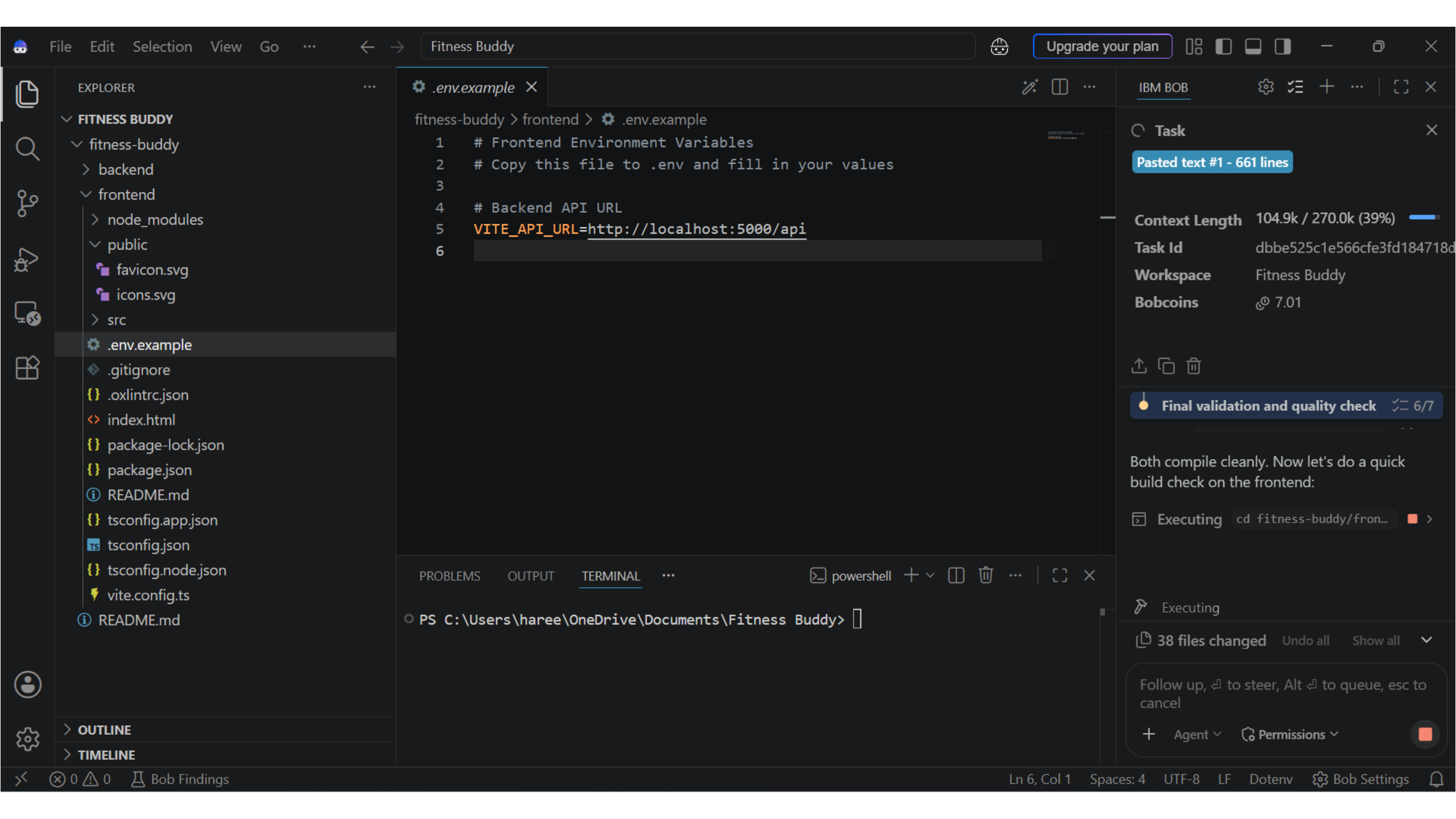
UTF-8

LF

Dotenv

Bob Settings

🔔



EXPLORER

FITNESS BUDDY

fitness-buddy

> backend

frontend

> node_modules

public

favicon.svg

icons.svg

> src

.env.example

.gitignore

{ } .oxlinttrc.json

< > index.html

{ } package-lock.json

{ } package.json

i README.md

{ } tsconfig.app.json

tsconfig.json

{ } tsconfig.node.json

vite.config.ts

i README.md

> OUTLINE

> TIMELINE

.env.example

fitness-buddy > frontend > .env.example

```
1 # Frontend Environment Variables
2 # Copy this file to .env and fill in your values
3
4 # Backend API URL
5 VITE_API_URL=http://localhost:5000/api
6
```

PROBLEMS

OUTPUT

TERMINAL

powershell

PS C:\Users\haree\OneDrive\Documents\Fitness Buddy>

IBM BOB

Task

Pasted text #1 - 661 lines

Context Length 104.9k / 270.0k (39%)

Task Id dbbe525c1e566cfe3fd184718d

Workspace Fitness Buddy

Bobcoins 7.01

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Executing cd fitness-buddy/fron...

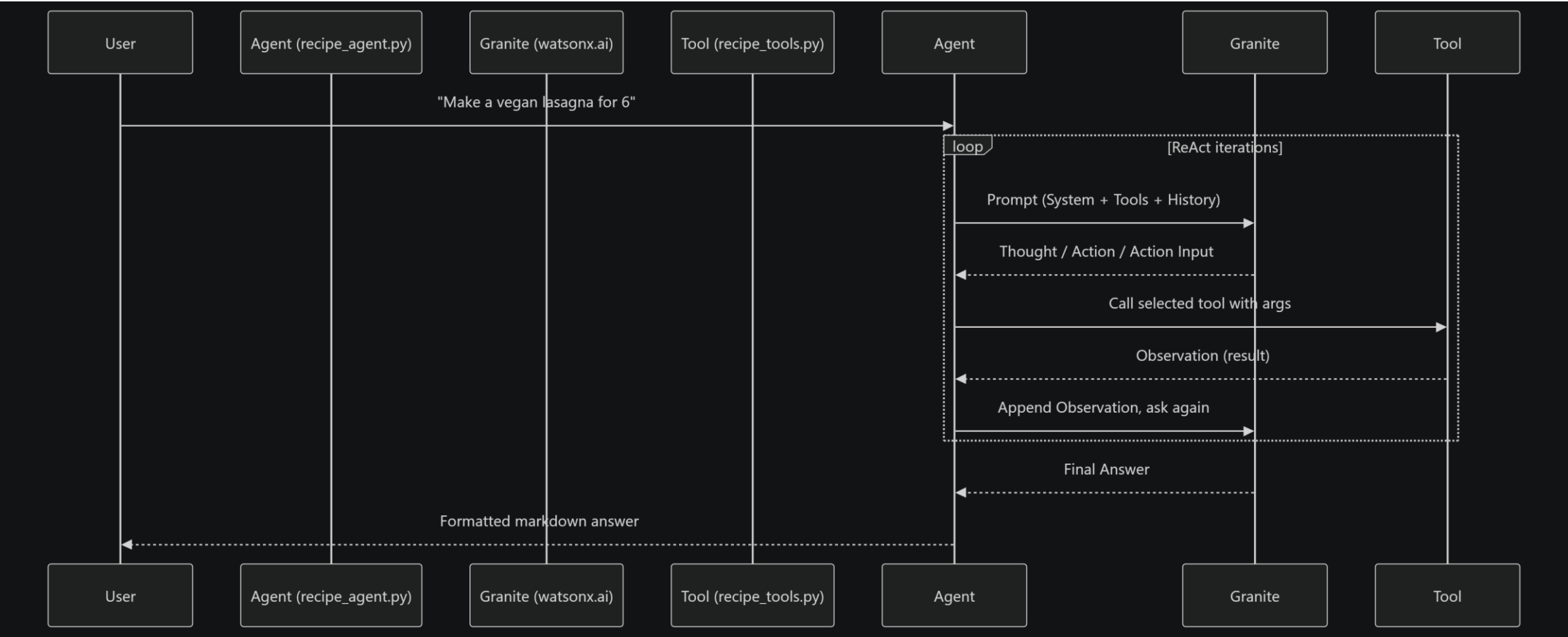
Executing

38 files changed Undo all Show all

Follow up, ↵ to steer, Alt ↵ to queue, esc to cancel

+ Agent Permissions

Architecture blueprint for IBM bob:



Role of Agentic AI in the solution

Role of Agentic AI in Fitness Buddy

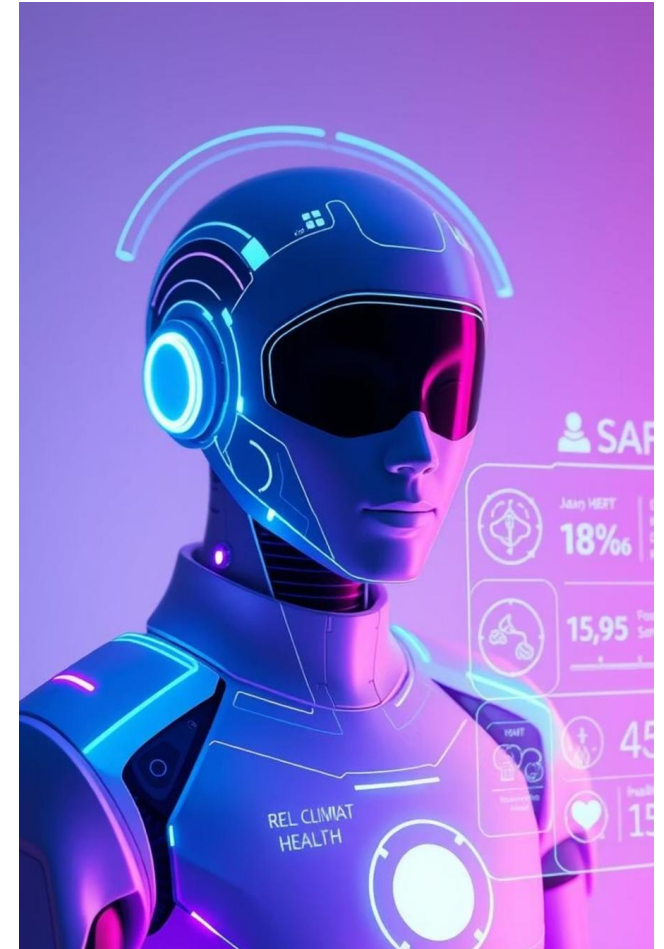
Agentic AI enables Fitness Buddy to function as an intelligent and autonomous fitness assistant rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized AI agents to provide personalized fitness and lifestyle guidance.

Each agent performs a specific role—such as generating workout plans, suggesting nutritious meals, providing fitness advice, and analyzing user habits—while working collaboratively. This **multi-agent approach** helps the system handle different tasks efficiently and provide relevant recommendations.

Agentic AI also provides **context awareness**, understanding user details such as fitness level, goals, available workout time, equipment, and dietary preferences to generate tailored recommendations. It supports **real-time adaptation**, allowing suggestions to change when the user provides new information.

Additionally, the agents can communicate and coordinate with each other, enabling better decision-making and more personalized results.

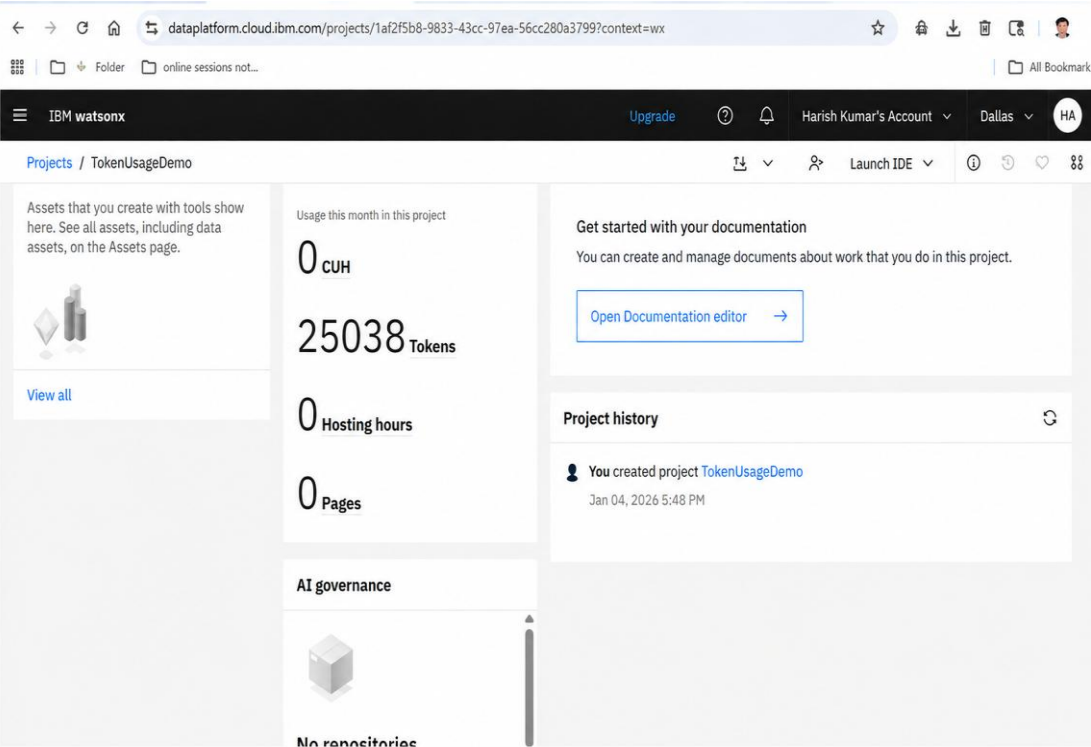
Overall, Agentic AI makes Fitness Buddy **proactive, personalized, and goal-driven**, acting like a **smart digital fitness coach** that helps users build healthy habits and maintain a healthier lifestyle.



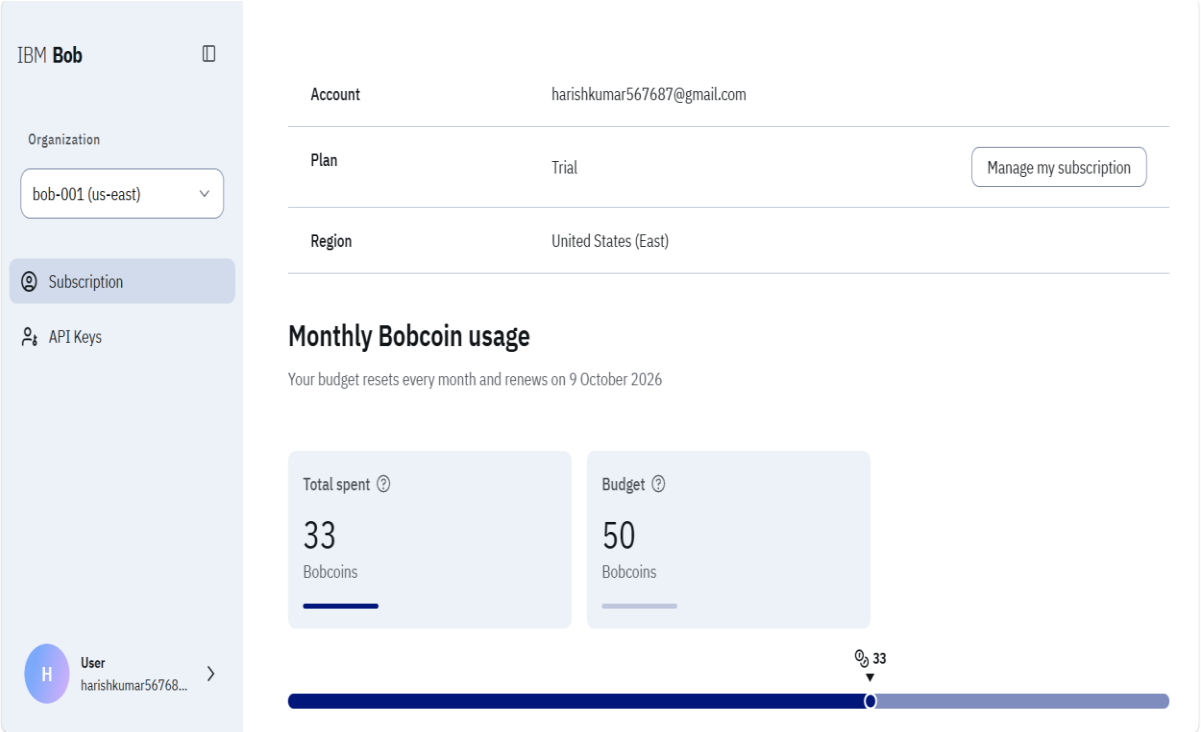
Token Usage

11,761 token are used

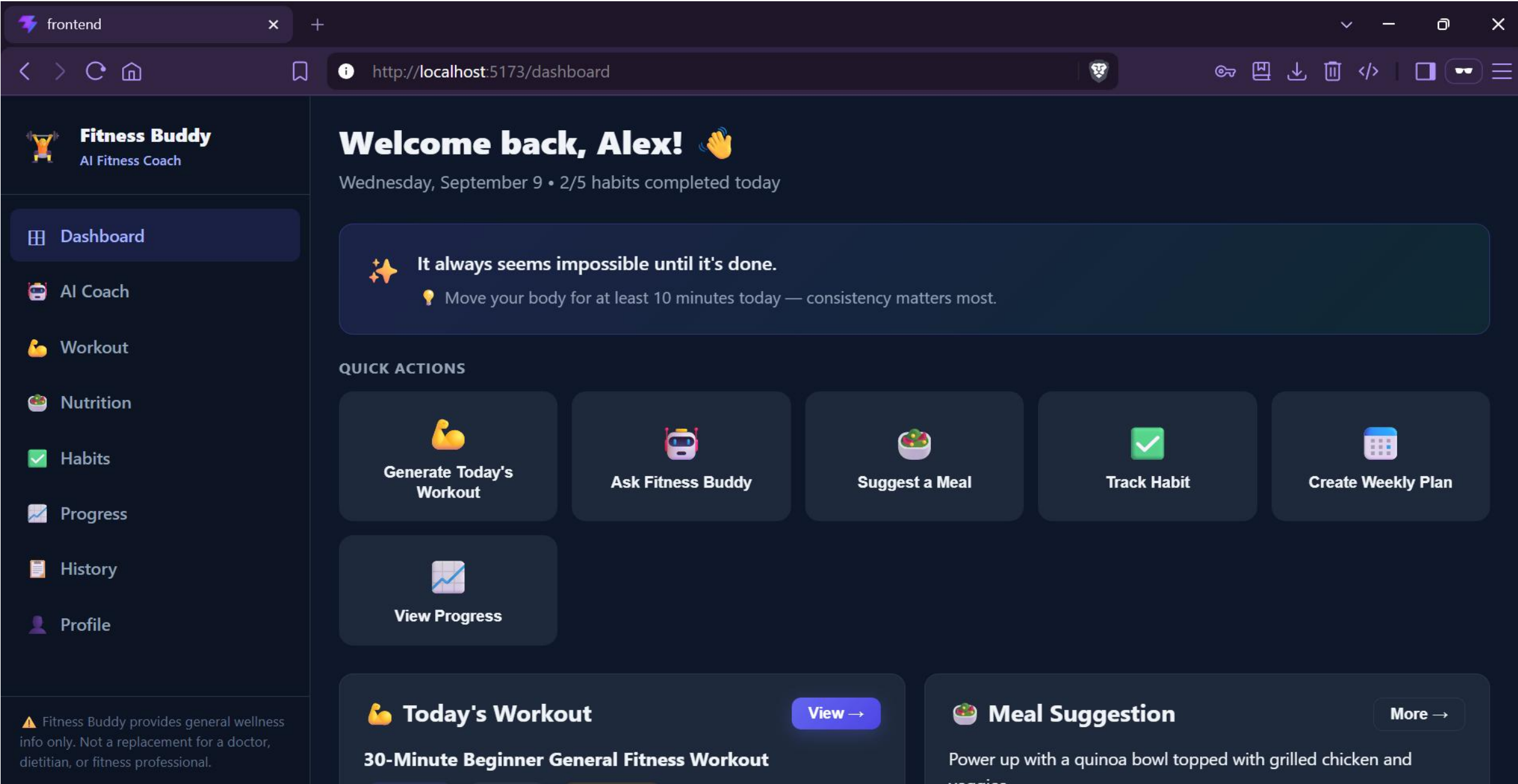
Before Prompt



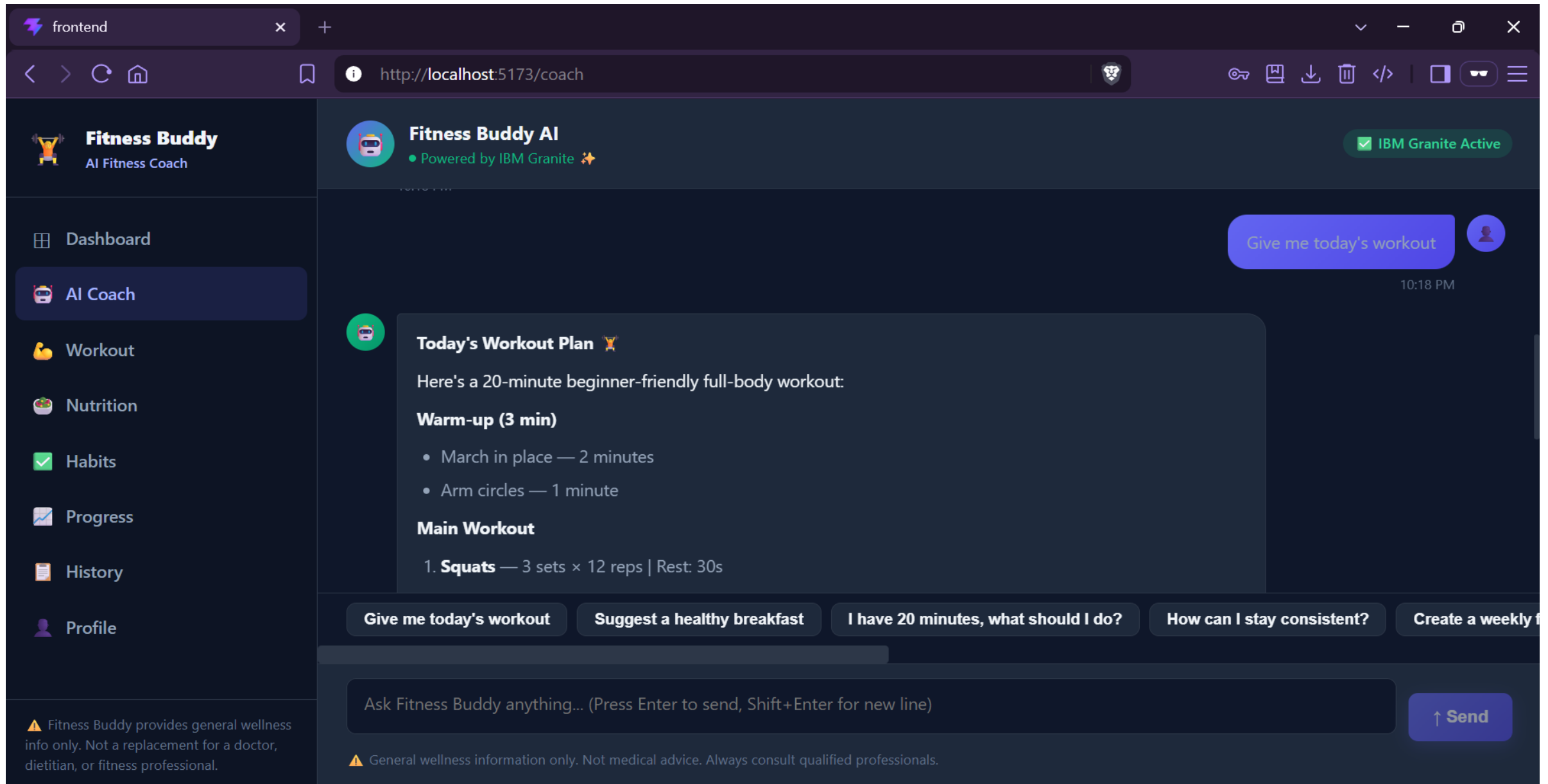
After the Prompt

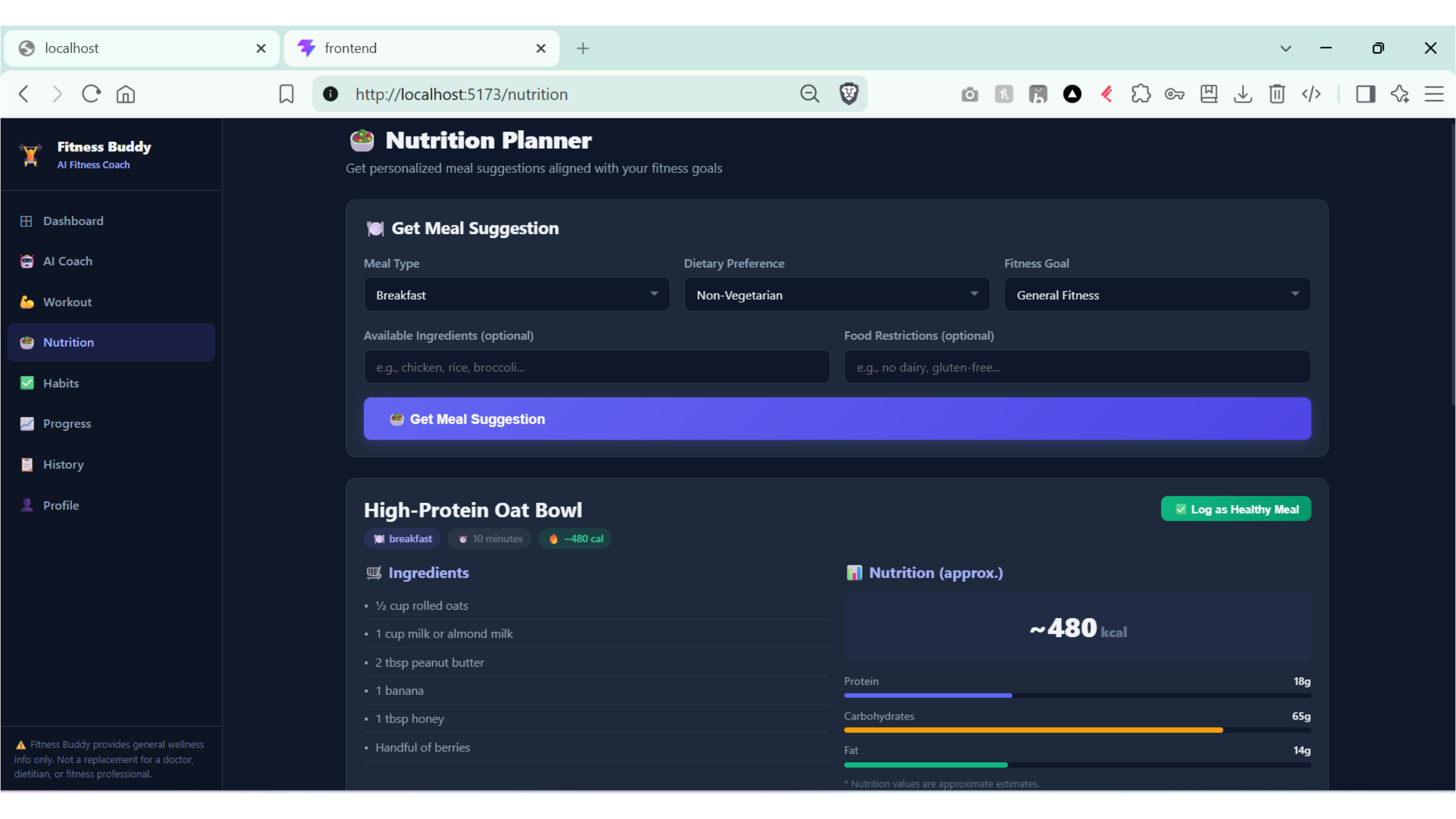


Project Output Screenshot



Project Output Screenshot





🥗 Nutrition Planner

Get personalized meal suggestions aligned with your fitness goals

🍽️ Get Meal Suggestion

Meal Type

Breakfast

Dietary Preference

Non-Vegetarian

Fitness Goal

General Fitness

Available Ingredients (optional)

e.g., chicken, rice, broccoli...

Food Restrictions (optional)

e.g., no dairy, gluten-free...

🥗 Get Meal Suggestion

High-Protein Oat Bowl

✅ Log as Healthy Meal

🍽️ breakfast ⌚ 10 minutes 🔥 ~480 cal

🛒 Ingredients

- ½ cup rolled oats
- 1 cup milk or almond milk
- 2 tbsp peanut butter
- 1 banana
- 1 tbsp honey
- Handful of berries

📊 Nutrition (approx.)

~480 kcal



* Nutrition values are approximate estimates.

Novelty and Uniqueness

The proposed **Fitness Buddy** stands out by combining **Agentic AI, RAG, and personalization** into a single intelligent fitness and wellness system.

- Multi-Agent Intelligence** – Multiple specialized agents work together to provide personalized workout, nutrition, and fitness guidance.
 - RAG-Based Accuracy** – Retrieves relevant information from reliable sources before generating AI responses, helping provide more trustworthy recommendations.
 - Hyper-Personalization** – Tailors workout plans, meal suggestions, and lifestyle guidance based on fitness goals, fitness level, available time, equipment, and dietary preferences.
 - Real-Time Feedback Loop** – Adapts recommendations based on user progress, completed workouts, habits, and new inputs.
 - Multi-Modal Input** – Can be extended to support text, voice, and image-based interaction for better usability.
 - Preventive Health Focus** – Encourages healthy lifestyle habits through fitness, nutrition, hydration, and consistency guidance.
 - Scalable & Enterprise-Ready** – Built using **IBM watsonx.ai and IBM Granite models**, providing a foundation for reliable AI deployment and governance.
- This makes the solution a **smart, adaptive, and proactive digital fitness coach**, rather than just a static workout recommendation tool.

Git Hub Link

- Please create public github repository (Enable Add README button while creating repository)

with format –

Github URL – <https://github.com/Hareesh0x01/Fitness-Buddy-IBM-Granite-AI.git>

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Future Scope

1. Integration with Wearables & Health Devices –

The Fitness Buddy system can be integrated with **smartwatches and fitness trackers** to collect real-time data such as steps, calories burned, heart rate, and sleep patterns. This will enable the AI assistant to provide more accurate and dynamic fitness, lifestyle, and nutrition recommendations based on actual user activity and health metrics.

2. AI-Powered Voice Assistant & Multilingual Support –

Enhancing Fitness Buddy with a **voice-enabled AI assistant and multilingual support** will improve accessibility. Users will be able to interact naturally through voice and in their preferred language, making the solution more inclusive and suitable for users from diverse linguistic backgrounds.

3. Personalized Health Insights –

In the future, the system can analyze long-term activity, workout, and habit data to provide **personalized health and fitness insights**, helping users understand their progress and improve their daily routines.

4. Advanced AI Fitness Planning –

The system can be enhanced to generate **dynamic weekly and monthly fitness plans** that automatically adapt according to the user's progress, consistency, goals, and changing preferences.

5. Integration with More Health Platforms –

Fitness Buddy can be connected with popular **health and fitness platforms** to synchronize activity and wellness data, creating a more complete picture of the user's fitness journey.

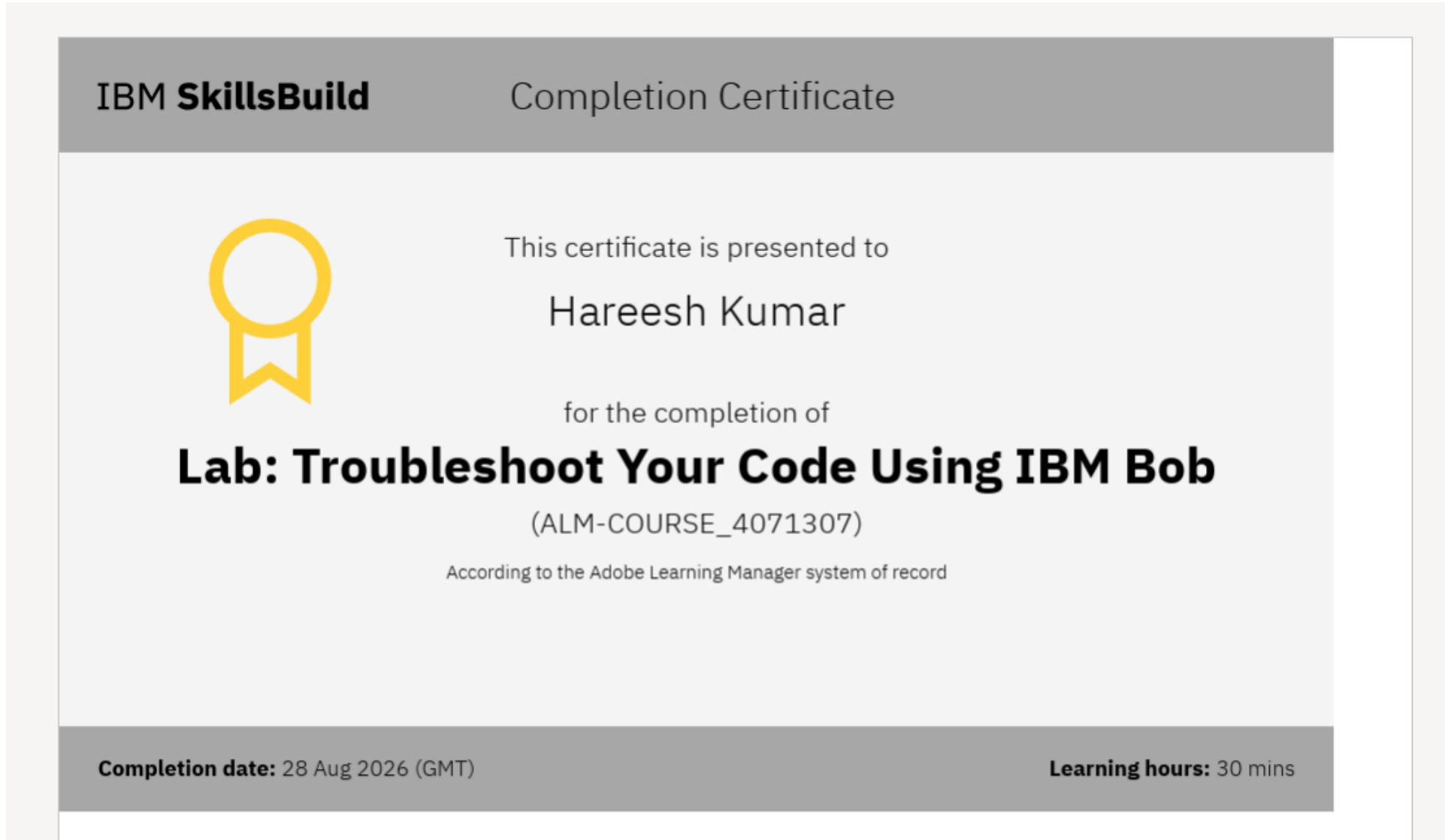
Certificates Earned:

Please add your credly certificate(Getting Started with Artificial Intelligence)



Certificates Earned:

Add your IBM BOB Certificate



Thank You!

Thank you for your time and interest.